university of groningen

Models and Semantics of Computation (MaSC)

Lecture 11: Strong and Weak Bisimilarity in CCS

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CCS: Syntax



$p, q ::= \mathrm{nil}$	inactive process
$\mid x$	process variable (for recursion)
$\mid \mu.p$	action prefix
$\mid p \backslash \alpha$	restricted channel
$\mid p[\phi]$	channel relabeling
$\mid p + q$	non deterministic choice
$\mid p \mid q$	parallel composition
$\mid \mathrm{rec}\, x.\, p$	recursion

(Note: operators are listed in order of precedence)

Structural Operational Semantics

act
$$\frac{}{\mu.p \xrightarrow{\mu} p}$$

SumL
$$\frac{p_1 \xrightarrow{\mu} q}{p_1 + p_2 \xrightarrow{\mu} q}$$

SumR
$$\frac{p_2 \xrightarrow{\mu} q}{p_1 + p_2 \xrightarrow{\mu} q}$$

Rec
$$\frac{p[\mathrm{rec}\, x.\, p/x] \xrightarrow{\mu} q}{\mathrm{rec}\, x.\, p \xrightarrow{\mu} q}$$

ParL
$$\frac{p_1 \xrightarrow{\mu} q_1}{p_1 \mid p_2 \xrightarrow{\mu} q_1 \mid p_2}$$

Com
$$\frac{p_1 \xrightarrow{\lambda} q_1 \qquad p_2 \xrightarrow{\overline{\lambda}} q_2}{p_1 \mid p_2 \xrightarrow{\tau} q_1 \mid q_2}$$

ParR
$$\frac{p_2 \xrightarrow{\mu} q_2}{p_1 \mid p_2 \xrightarrow{\mu} p_1 \mid q_2}$$

Res
$$\frac{p \xrightarrow{\mu} q \qquad \mu \notin \{\alpha, \overline{\alpha}\}}{p \backslash \alpha \xrightarrow{\mu} q \backslash \alpha}$$

Rel
$$\frac{p \xrightarrow{\mu} q}{p[\phi] \xrightarrow{\phi(\mu)} q[\phi]}$$



Strong Bisimilarity

Strong Bisimulation



Consider

- A set of processes $\mathcal{P}$.
- A binary relation $\mathcal{R} \subseteq \mathcal{P} \times \mathcal{P}$. We write $p \; \mathcal{R} \; q$ when $(p, q) \in \mathcal{R}$.

$\mathcal{R}$ is a strong bisimulation if and only if

$$\forall p, q. (p, q) \in \mathcal{R} \Rightarrow \begin{cases} \forall \mu, p'. p \xrightarrow{\mu} p' & \Rightarrow \exists q'. q \xrightarrow{\mu} q' \wedge p' \; \mathcal{R} \; q' \\ \quad \wedge & \\ \forall \mu, q'. q \xrightarrow{\mu} q' & \Rightarrow \exists p'. p \xrightarrow{\mu} p' \wedge p' \; \mathcal{R} \; q' \end{cases}$$

Strong Bisimulation



Consider

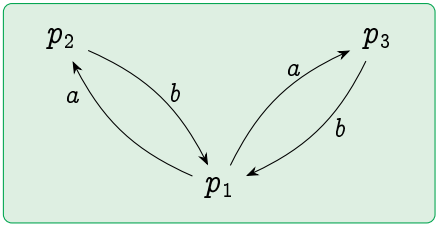
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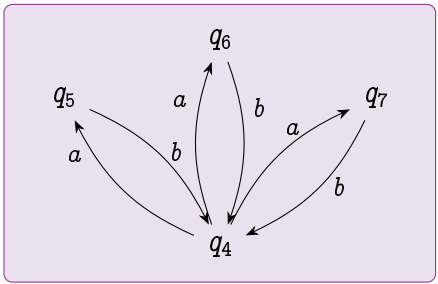
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$$\forall p, q.\, (p, q) \in \mathcal{R} \Rightarrow \begin{cases} \forall \mu, p'.\, p \xrightarrow{\mu} p' & \Rightarrow \exists q'.\, q \xrightarrow{\mu} q' \wedge p' \mathcal{R} q' \\ \quad \wedge & \\ \forall \mu, q'.\, q \xrightarrow{\mu} q' & \Rightarrow \exists p'.\, p \xrightarrow{\mu} p' \wedge p' \mathcal{R} q' \end{cases}$$

Intuition: If two processes are related, then for any move of Alice, Bob can find a move that leads to related processes. That is, Bob has a *winning strategy*.

Exercise





Define the relation

$$\mathcal{R} = \{(p_1, q_4), (p_2, q_5), (p_3, q_5), (p_2, q_6), (p_2, q_7)\}$$

Is $\mathcal{R}$ a strong bisimulation?

Properties of Strong Bisimulation (1)



Union:
If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two strong bisimulations then $\mathcal{R}_1 \cup \mathcal{R}_2$ is a strong bisimulation.

Proof:

Properties of Strong Bisimulation (1)



Union:
If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two strong bisimulations then $\mathcal{R}_1 \cup \mathcal{R}_2$ is a strong bisimulation.

Proof:

- Take $(p, q) \in \mathcal{R}_1 \cup \mathcal{R}_2$.
  First suppose $p \xrightarrow{\mu} p'$. We must find $q \xrightarrow{\mu} q'$ such that $(p', q') \in \mathcal{R}_1 \cup \mathcal{R}_2$.

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  First suppose $p \xrightarrow{\mu} p'$. We must find $q \xrightarrow{\mu} q'$ such that $(p', q') \in \mathcal{R}_1 \cup \mathcal{R}_2$.
- Since $(p, q) \in \mathcal{R}_1 \cup \mathcal{R}_2$ we know that $p \, \mathcal{R}_i \, q$ for some $i \in \{1, 2\}$.
  Since $\mathcal{R}_i$ is a strong bisimulation and $p \xrightarrow{\mu} p'$, then $q \xrightarrow{\mu} q'$ and $p' \, \mathcal{R}_i \, q'$.
  Hence $(p', q') \in \mathcal{R}_1 \cup \mathcal{R}_2$.

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  Since $\mathcal{R}_i$ is a strong bisimulation and $p \xrightarrow{\mu} p'$, then $q \xrightarrow{\mu} q'$ and $p' \; \mathcal{R}_i \; q'$.
  Hence $(p', q') \in \mathcal{R}_1 \cup \mathcal{R}_2$.
- Now suppose $q \xrightarrow{\mu} q'$. We must find $p \xrightarrow{\mu} p'$ such that $(p', q') \in \mathcal{R}_1 \cup \mathcal{R}_2$.
  The analysis is exactly as in the previous case.

Properties of Strong Bisimulation (2)



Inverse:
If $\mathcal{R}$ is a strong bisimulation then
$\mathcal{R}^{-1} \triangleq \{(q, p) \mid (p, q) \in \mathcal{R}\}$ is a strong bisimulation.

Proof:

Properties of Strong Bisimulation (2)



Inverse:
If $\mathcal{R}$ is a strong bisimulation then
$\mathcal{R}^{-1} \triangleq \{(q, p) \mid (p, q) \in \mathcal{R}\}$ is a strong bisimulation.

Proof:

- Take $(q, p) \in \mathcal{R}^{-1}$.
  First suppose $q \xrightarrow{\mu} q'$. We must find $p \xrightarrow{\mu} p'$ such that $(q', p') \in \mathcal{R}^{-1}$.
- Since $(q, p) \in \mathcal{R}^{-1}$ we know that $p \mathcal{R} q$. Since $\mathcal{R}$ is a strong bisimulation and $q \xrightarrow{\mu} q'$, then $p \xrightarrow{\mu} p'$ and $p' \mathcal{R} q'$. Hence $(q', p') \in \mathcal{R}^{-1}$.

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- Since $(q, p) \in \mathcal{R}^{-1}$ we know that $p \mathcal{R} q$. Since $\mathcal{R}$ is a strong bisimulation and $q \xrightarrow{\mu} q'$, then $p \xrightarrow{\mu} p'$ and $p' \mathcal{R} q'$. Hence $(q', p') \in \mathcal{R}^{-1}$.
- Now suppose $p \xrightarrow{\mu} p'$. We must find $q \xrightarrow{\mu} q'$ such that $(q', p') \in \mathcal{R}^{-1}$. The analysis is exactly as in the previous case.

Properties of Strong Bisimulation (3)



Composition:
If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two strong bisimulations then
$\mathcal{R}_2 \circ \mathcal{R}_1 = \{(p, q) \mid \exists r \,.\, p \;\mathcal{R}_1\; r \wedge r \;\mathcal{R}_2\; q\}$ is a strong bisimulation.

Proof:

Properties of Strong Bisimulation (3)



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If $\mathcal{R}_1$ and $\mathcal{R}_2$ are two strong bisimulations then
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Proof:

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  First suppose $p \xrightarrow{\mu} p'$. We must find $q \xrightarrow{\mu} q'$ such that $(p', q') \in \mathcal{R}_2 \circ \mathcal{R}_1$.
- Since $(p, q) \in \mathcal{R}_2 \circ \mathcal{R}_1$ we know that $p\ \mathcal{R}_1\ r$ and $r\ \mathcal{R}_2\ q$, for some r.
  Since $\mathcal{R}_1$ is a strong bisimulation and $p \xrightarrow{\mu} p'$, then $r \xrightarrow{\mu} r'$ and $p'\ \mathcal{R}_1\ r'$.

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  Since $\mathcal{R}_1$ is a strong bisimulation and $p \xrightarrow{\mu} p'$, then $r \xrightarrow{\mu} r'$ and $p'\ \mathcal{R}_1\ r'$.
- Also, since $\mathcal{R}_2$ is a strong bisimulation and $r \xrightarrow{\mu} r'$, then $q \xrightarrow{\mu} q'$ and $r'\ \mathcal{R}_1\ q'$. Hence $(p', q') \in \mathcal{R}_2 \circ \mathcal{R}_1$.

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- Since $(p, q) \in \mathcal{R}_2 \circ \mathcal{R}_1$ we know that $p\ \mathcal{R}_1\ r$ and $r\ \mathcal{R}_2\ q$, for some r.
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- Now suppose $q \xrightarrow{\mu} q'$. We must find $p \xrightarrow{\mu} p'$ such that $(p', q') \in \mathcal{R}_2 \circ \mathcal{R}_1$.
  The analysis is exactly as in the previous case.

Strong Bisimilarity ($\simeq$)

$p \simeq q$ iff exists a strong bisimulation $\mathcal{R}$ with $(p, q) \in \mathcal{R}$

$$\simeq \triangleq \bigcup_{\mathcal{R} \text{ is sb}} \mathcal{R}$$

Properties:

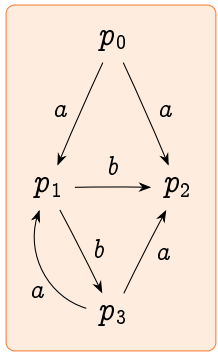
- Strong bisimilarity is an equivalence relation (reflexive, symmetric, transitive).
- Strong bisimilarity is a strong bisimulation.
- Strong bisimilarity is the largest strong bisimulation.

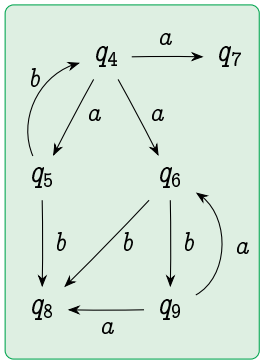
Example: Constructing a Bisimulation

Q: How to prove $p_0 \simeq q_4$?

A: Show a bisimulation containing (p_0, q_4).

- $\mathcal{R} = \{(p_0, q_4)\}$



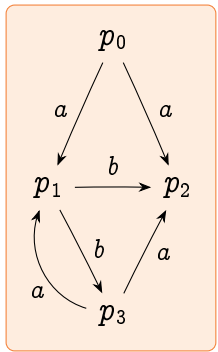


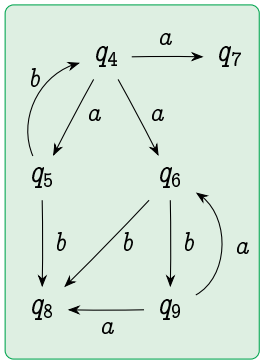
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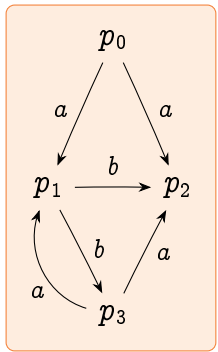


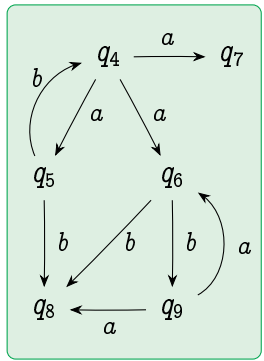
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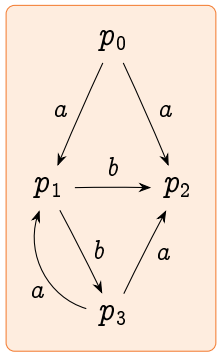


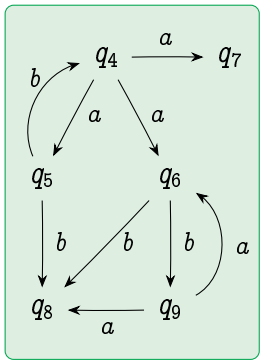
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- For any $q_4 \xrightarrow{\lambda} q'$, we must find a $p_0 \xrightarrow{\lambda} p'$ so that $(p', q') \in \mathcal{R}$.
- Deadlocked states (no outgoing transitions): p_2, q_7, q_8.



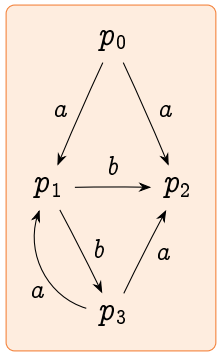


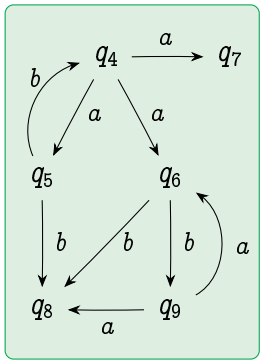
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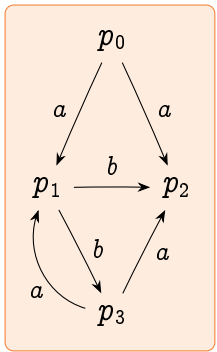


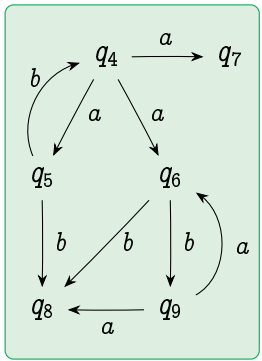
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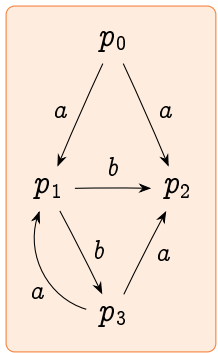


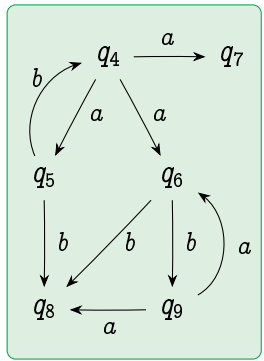
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  All a-moves from q_4: q_5, q_6, q_7.
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  We don't add (p_1, q_7).



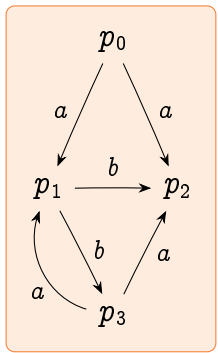


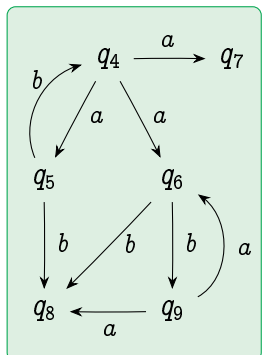
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- $\mathcal{R} = \{(p_0, q_4)\} \cup \{(p_1, q_5), (p_1, q_6)\} \cup \{(p_2, q_7)\}$
- We have $p_0 \xrightarrow{a} p_1$.
  All a-moves from q_4: q_5, q_6, q_7.
- We add (p_1, q_5) and (p_1, q_6) to $\mathcal{R}$.
  We don't add (p_1, q_7).
- We also have $p_0 \xrightarrow{a} p_2$. Given all a-moves from q_4, we add (p_2, q_7).



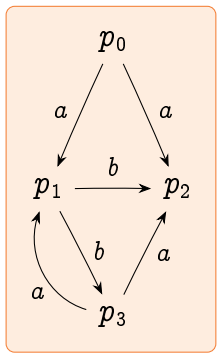


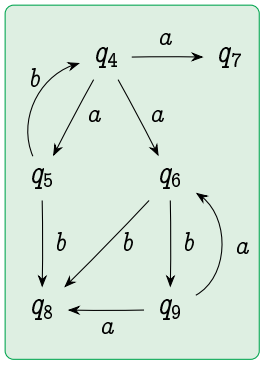
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- We have $p_0 \xrightarrow{a} p_1$.
  All a-moves from q_4: q_5, q_6, q_7.
- We add (p_1, q_5) and (p_1, q_6) to $\mathcal{R}$.
  We don't add (p_1, q_7).
- We also have $p_0 \xrightarrow{a} p_2$. Given all a-moves from q_4, we add (p_2, q_7).
- We now check all q' such that $q_4 \xrightarrow{a} q'$: all pairs are in $\mathcal{R}$.



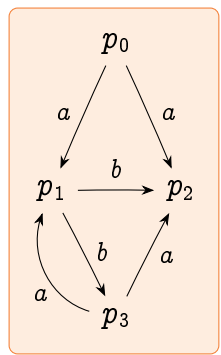


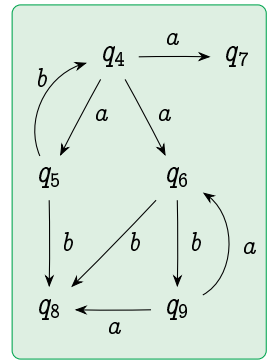
Example: Constructing a Bisimulation

Q: How to prove $p_0 \simeq q_4$?

A: Show a bisimulation containing (p_0, q_4).

- $\mathcal{R} = \{(p_0, q_4)\} \cup \{(p_1, q_5), (p_1, q_6)\}$
  $\cup \{(p_2, q_7)\}$
- We check (p_1, q_5).
  All b-moves from p_1: p_2, p_3.
  All b-moves from q_5: q_4, q_8.
  From p_1, we add $(p_2, q_8), (p_3, q_4)$ to $\mathcal{R}$.
  From q_5, no new pairs.



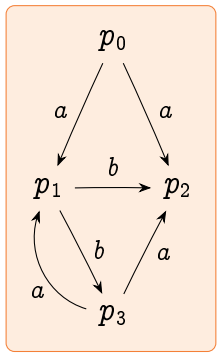


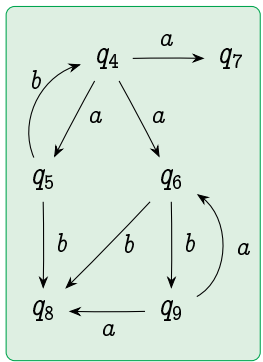
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- We check (p_2, q_7) and (p_2, q_8): they satisfy vacuously the conditions.



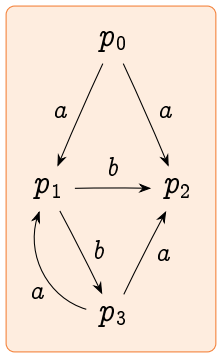


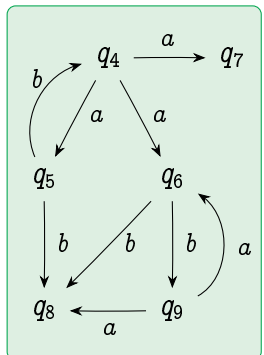
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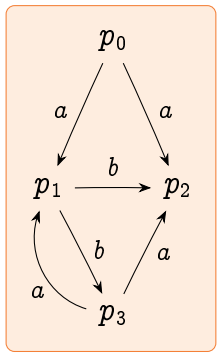


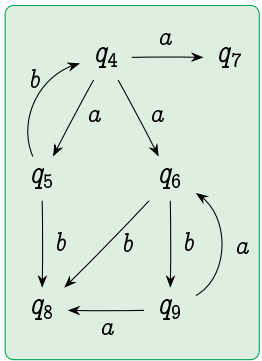
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  All a-moves from q_4: q_5, q_6, q_7.
  No new pairs.



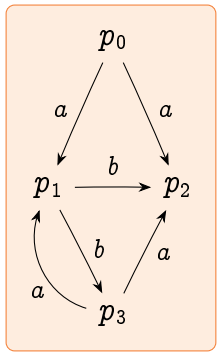


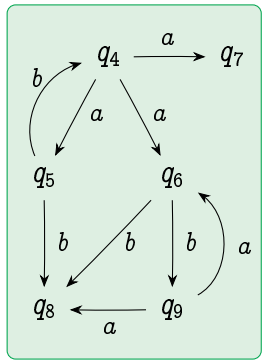
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  $\cup \{(p_2, q_8), (p_3, q_4)\} \cup \{(p_3, q_9)\}$
- Finally, we check (p_3, q_9).
  No new pairs and we are done.





Strong Bisimilarity is a Congruence



Congruence allows us to replace any process with an equivalent one in any context, preserving the overall behavior.

Congruence: For all contexts $\mathbb{C}$, $p \simeq q$ implies $\mathbb{C}[p] \simeq \mathbb{C}[q]$.

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More precisely:

- $p \simeq q \Rightarrow \mu.p \simeq \mu.q$
- $p \simeq q \Rightarrow p\backslash\alpha \simeq q\backslash\alpha$
- $p \simeq q \Rightarrow p[\phi] \simeq q[\phi]$
- $p_0 \simeq q_0 \wedge p_1 \simeq q_1 \Rightarrow p_0 + p_1 \simeq q_0 + q_1$
- $p_0 \simeq q_0 \wedge p_1 \simeq q_1 \Rightarrow p_0 \mid p_1 \simeq q_0 \mid q_1$

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In all cases, the proof technique consists in:

(i) devising a relation $\mathcal{R}$ large enough to contain all pairs of interest; and
(ii) showing that $\mathcal{R}$ is contained in strong bisimilarity.

Some Laws

Properties of sum and parallel composition:

$$p + \mathrm{nil} \simeq p \qquad p \mid \mathrm{nil} \simeq p$$

$$p + q \simeq q + p \qquad p \mid q \simeq q \mid p$$

$$p + (q + r) \simeq (p + q) + r \qquad p \mid (q \mid r) \simeq (p \mid q) \mid r$$

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Properties of restriction and relabeling:

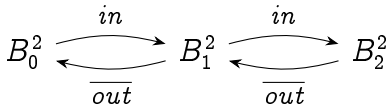
$$\mathrm{nil}\backslash\alpha \simeq \mathrm{nil} \qquad \mathrm{nil}[\phi] \simeq \mathrm{nil}$$
$$(\mu.p)\backslash\alpha \simeq \mathrm{nil} \quad \text{if } \mu \in \{\alpha, \overline{\alpha}\} \qquad (\mu.p)[\phi] \simeq \phi(\mu).p[\phi]$$
$$(\mu.p)\backslash\alpha \simeq \mu.(p\backslash\alpha) \quad \text{if } \mu \notin \{\alpha, \overline{\alpha}\}$$
$$(p + q)\backslash\alpha \simeq (p\backslash\alpha) + (q\backslash\alpha) \qquad (p + q)[\phi] \simeq (p[\phi]) + (q[\phi])$$
$$p\backslash\alpha\backslash\alpha \simeq p\backslash\alpha \qquad p[\phi][\eta] \simeq p[\eta \circ \phi]$$
$$p\backslash\alpha\backslash\beta \simeq p\backslash\beta\backslash\alpha$$



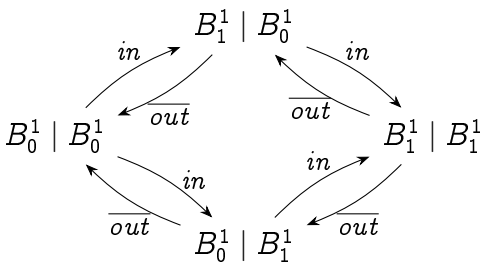
Weak Transitions

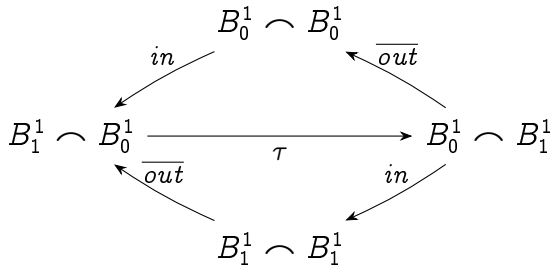
Comparing Buffers

- The sequential buffer of capacity 2:



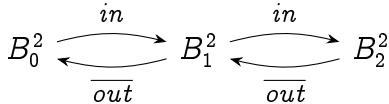
- The parallel buffer and the linked buffer:



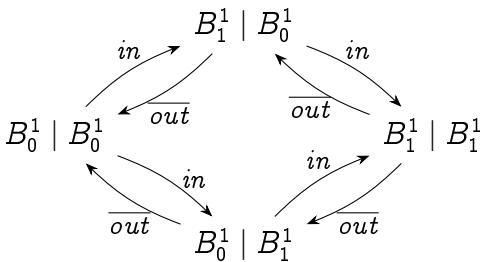


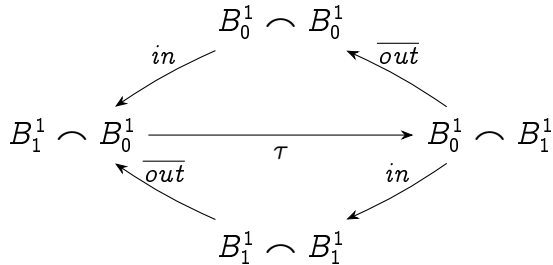
Comparing Buffers

- The sequential buffer of capacity 2:



- The parallel buffer and the linked buffer:





- Are they all strongly bisimilar?

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  - with a concrete implementation (many τ's)
- Can we abstract away from them? Can we find a broader equivalence?

Weak Transitions

$$p \stackrel{\tau}{\Longrightarrow} q \text{ iff } p \ (\stackrel{\tau}{\longrightarrow})^* \ q$$

$$(p = q) \vee p \stackrel{\tau}{\longrightarrow} \cdots \stackrel{\tau}{\longrightarrow} q$$

$$p \stackrel{\lambda}{\Longrightarrow} q \text{ iff } \exists p', q' . \, p \stackrel{\tau}{\Longrightarrow} p' \stackrel{\lambda}{\longrightarrow} q' \stackrel{\tau}{\Longrightarrow} q$$

Intuition:

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- $p \overset{\lambda}{\Longrightarrow} q$ means: p can reach q via a λ-transition possibly preceded and followed by empty/finite sequences of τ-transitions.

Weak Bisimulation Game

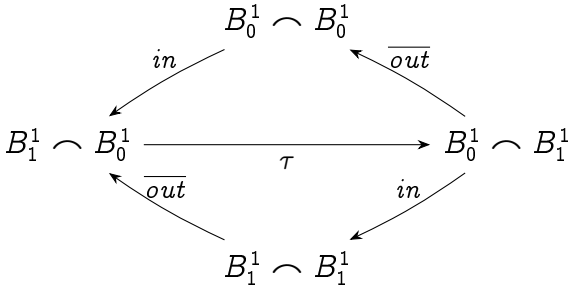


More power to the defender! At each turn:

- **Alice** chooses one process and one ordinary (strong) transitions.
- **Bob** must reply with a weak transition of the other process, matching the label of the transition chosen by Alice.

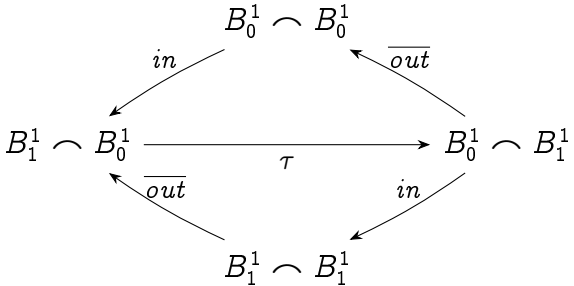
If Alice picks a silent transition, Bob can just leave the other process idle—he can choose not to move.

Example



We have:

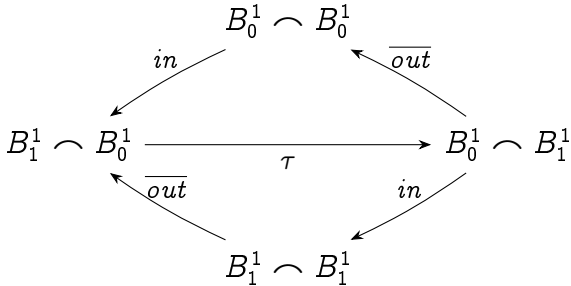
Example



We have:

- $B_0^1 \frown B_0^1 \stackrel{\tau}{\Longrightarrow} B_0^1 \frown B_0^1$

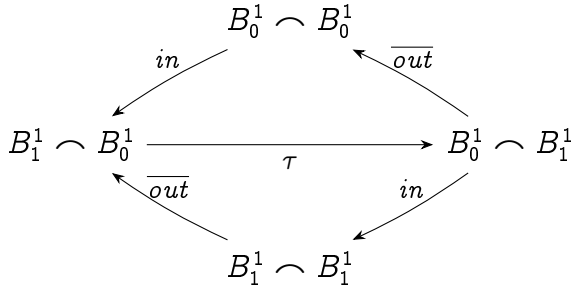
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- $B_1^1 \frown B_0^1 \stackrel{\overline{out}}{\Longrightarrow} B_0^1 \frown B_0^1$

Weak Bisimulation

$\mathcal{R}$ is a weak bisimulation if and only if

$$\forall p, q.\, (p, q) \in \mathcal{R} \Rightarrow \begin{cases} \forall \mu, p'.\, p \xrightarrow{\mu} p' & \Rightarrow \exists q'.\, q \stackrel{\mu}{\Longrightarrow} q' \wedge p' \,\mathcal{R}\, q' \\ \quad \wedge & \\ \forall \mu, q'.\, q \xrightarrow{\mu} q' & \Rightarrow \exists p'.\, p \stackrel{\mu}{\Longrightarrow} p' \wedge p' \,\mathcal{R}\, q' \end{cases}$$

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Weak Bisimilarity ($\approx$)

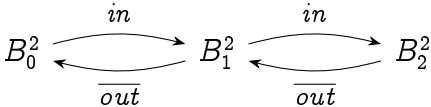


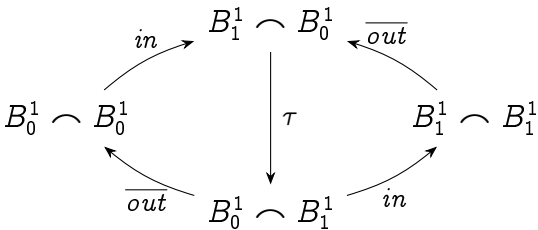
$p \approx q$ iff exists a weak bisimulation $\mathcal{R}$ with $(p, q) \in \mathcal{R}$

Properties:

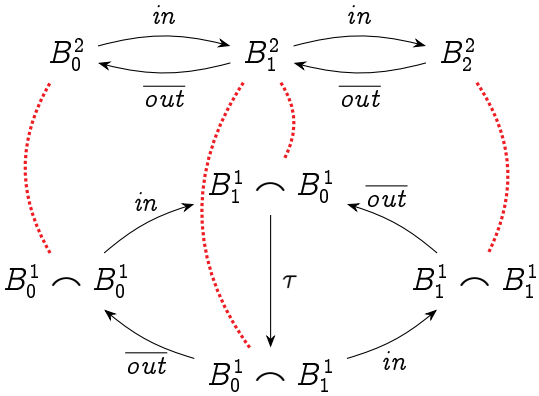
- Weak bisimilarity is an equivalence relation (reflexive, symmetric, transitive).
- Any strong bisimulation is a weak bisimulation.
- Strong bisimilarity implies weak bisimilarity.

Example


B_0^2
in
$\overline{out}$
B_1^2
in
$\overline{out}$
B_2^2


$B_1^1 \frown B_0^1$
in
$\overline{out}$
$B_0^1 \frown B_0^1$
τ
$B_1^1 \frown B_1^1$
$\overline{out}$
$B_0^1 \frown B_1^1$
in

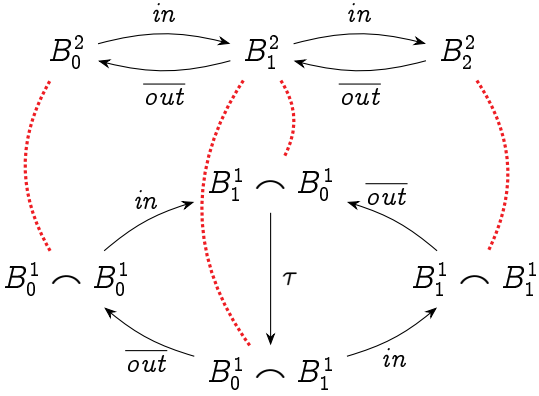
Example



We have the weak bisimulation:

$$\mathcal{R} \triangleq \{(B_0^2,\ B_0^1 \frown B_0^1),\ (B_1^2,\ B_0^1 \frown B_1^1),\ (B_1^2,\ B_1^1 \frown B_0^1),\ (B_2^2,\ B_1^1 \frown B_1^1)\}$$

Example



We have the weak bisimulation:

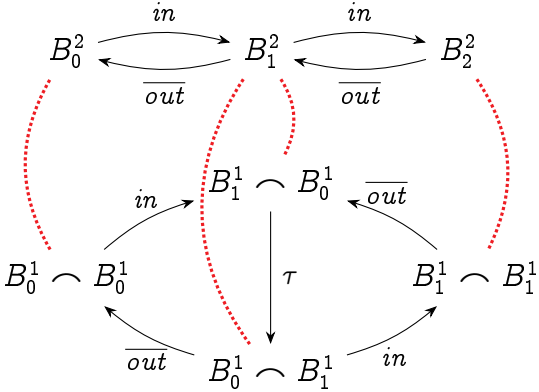
$$\mathcal{R} \triangleq \{(B_0^2,\ B_0^1 \frown B_0^1),\ (B_1^2,\ B_0^1 \frown B_1^1),\ (B_1^2,\ B_1^1 \frown B_0^1),\ (B_2^2,\ B_1^1 \frown B_1^1)\}$$

Some checks (1):

$$\begin{array}{ccc} B_0^2 & \xrightarrow{in} & B_1^2 \\ \mathcal{R} & & \mathcal{R} \\ B_0^1 \frown B_0^1 & \stackrel{in}{\Longrightarrow} & B_1^1 \frown B_0^1 \end{array}$$

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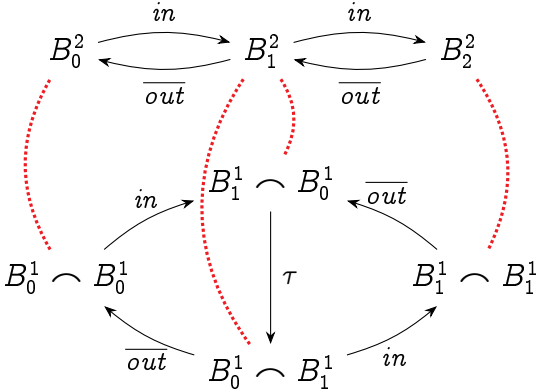
$$\mathcal{R} \triangleq \{(B_0^2,\ B_0^1 \frown B_0^1),\ (B_1^2,\ B_0^1 \frown B_1^1),\ (B_1^2,\ B_1^1 \frown B_0^1),\ (B_2^2,\ B_1^1 \frown B_1^1)\}$$

Some checks (2):

$$\begin{array}{ccc} B_1^2 & \xrightarrow{in} & B_2^2 \\ \mathcal{R} & & \mathcal{R} \\ B_1^1 \frown B_0^1 & \xRightarrow{in} & B_1^1 \frown B_1^1 \end{array}$$

$$\begin{array}{ccc} B_1^2 & \xrightarrow{\overline{out}} & B_0^2 \\ \mathcal{R} & & \mathcal{R} \\ B_1^1 \frown B_0^1 & \xRightarrow{\overline{out}} & B_0^1 \frown B_0^1 \end{array}$$

Example



We have the weak bisimulation:

$$\mathcal{R} \triangleq \{(B_0^2,\ B_0^1 \frown B_0^1),\ (B_1^2,\ B_0^1 \frown B_1^1),\ (B_1^2,\ B_1^1 \frown B_0^1),\ (B_2^2,\ B_1^1 \frown B_1^1)\}$$

Some checks (3):

$$\begin{array}{ccc} B_1^1 \frown B_0^1 & \xrightarrow{\tau} & B_0^1 \frown B_1^1 \\ \mathcal{R} & & \mathcal{R} \\ B_1^2 & \stackrel{\tau}{\Longrightarrow} & B_1^2 \end{array}$$

$$\begin{array}{ccc} B_2^2 & \xrightarrow{\overline{out}} & B_1^2 \\ \mathcal{R} & & \mathcal{R} \\ B_1^1 \frown B_1^1 & \stackrel{\overline{out}}{\Longrightarrow} & B_1^1 \frown B_0^1 \end{array}$$

Shortcomings of Weak Bisimilarity (1)



Under weak steps, guarded processes can have infinitely branching LTSs.

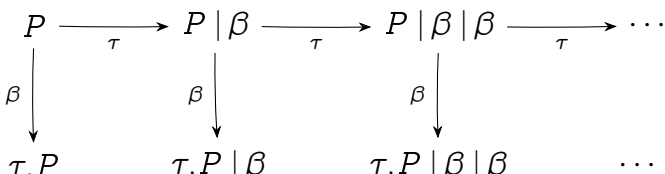
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Example

Consider $P \triangleq \mathrm{rec}\, x.\, \tau.x \mid \beta$. We have:



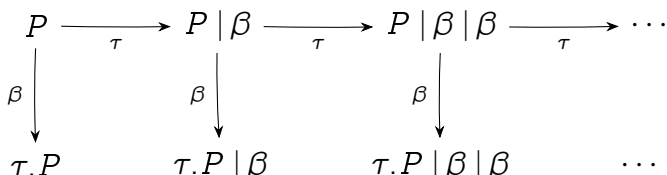
Shortcomings of Weak Bisimilarity (1)



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Note: τ-prefixes are not the issue. Consider $P \triangleq \mathrm{rec}\, x.\, (\alpha.x \mid \overline{\alpha} \mid \beta)\backslash\alpha$.

Shortcomings of Weak Bisimilarity (2)



Weak bisimilarity cannot distinguish between deadlock and silent divergence.

Example

- Consider the processes $\mathrm{rec}\, x.\, \tau.x$ and nil.
- We have

$$\mathrm{rec}\, x.\, \tau.x \approx \mathrm{nil}$$

because $\mathrm{rec}\, x.\, \tau.x \xrightarrow{\tau} \mathrm{rec}\, x.\, \tau.x$ and $\mathrm{nil} \stackrel{\tau}{\Longrightarrow} \mathrm{nil}$.

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Weak bisimilarity is not a congruence (with respect to $+$).

Example ($P \approx Q$ but $\mathbb{C}[P] \not\approx \mathbb{C}[Q]$)

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- Consider $P \triangleq \alpha$ and $Q \triangleq \tau.\alpha$. We have $P \approx Q$ because:
  - if $P \xrightarrow{\alpha} \mathbf{nil}$ then $Q \stackrel{\alpha}{\Longrightarrow} \mathbf{nil}$ and if $Q \xrightarrow{\tau} \alpha$ then $P \stackrel{\tau}{\Longrightarrow} P$.

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- A bisimulation game between $\mathbb{C}[P]$ and $\mathbb{C}[Q]$:

  Alice plays $\mathbb{C}[Q] \xrightarrow{\tau} \alpha$ (β is discarded) **Bob** can only reply $\mathbb{C}[P] \stackrel{\tau}{\Longrightarrow} \mathbb{C}[P]$

  Alice plays $\mathbb{C}[P] \xrightarrow{\beta} \mathrm{nil}$ **Bob** cannot reply: $\alpha \stackrel{\beta}{\not\Longrightarrow}$

  Alice wins: $\mathbb{C}[P] \not\approx \mathbb{C}[Q]$

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What to do?

Weak Observational Equivalence ($\approx\!\!\!\!\!\!\approx$)

$$p \approx\!\!\!\!\!\!\approx q \text{ iff } p \approx q \wedge \forall r.\ p + r \approx q + r$$

Equivalently:

$$p \approx\!\!\!\!\!\!\approx q \text{ iff } \begin{cases} \forall p'.\ p \xrightarrow{\tau} p' & \Rightarrow \exists q', q''.\ q \xrightarrow{\tau} q'' \stackrel{\tau}{\Longrightarrow} q' \wedge p' \approx q' \\ \forall p', \lambda.\ p \xrightarrow{\lambda} p' & \Rightarrow \exists q'.\ q \stackrel{\lambda}{\Longrightarrow} q' \wedge p' \approx q' \end{cases}$$

(and vice-versa for actions from q.)

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  An internal action $p \xrightarrow{\tau} p'$ must be matched by at least one internal action.

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  An internal action $p \xrightarrow{\tau} p'$ must be matched by at least one internal action.
- In the bisimulation game, Bob is not allowed to use an idle move at the very first turn; in subsequent turns, the ordinary weak bisimulation game applies.
- **Theorem:** $\approx\!\!\!\!\!\approx$ is the largest congruence contained in $\approx$.

Weak Observational Equivalence ($\cong$)

$$p \cong q \text{ iff } p \approx q \wedge \forall r.\ p + r \approx q + r$$

Example

- Consider processes

$$p' \triangleq \tau.\alpha \qquad q' \triangleq \alpha$$

  We have $p' \approx q'$ but also $p' \not\cong q'$.

- Let $p \triangleq \beta.p'$ and $q \triangleq \beta.q'$. We have both $p \approx q$ and $p \cong q$.

Weak Observational Equivalence ($\approxeq$)

$$p \approxeq q \text{ iff } p \approx q \wedge \forall r.\ p + r \approx q + r$$

Example

- Consider processes

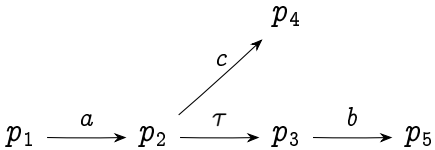
$$p' \triangleq \tau.\alpha \qquad q' \triangleq \alpha$$

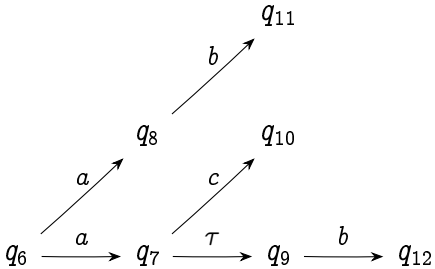
  We have $p' \approx q'$ but also $p' \not\approxeq q'$.
- Let $p \triangleq \beta.p'$ and $q \triangleq \beta.q'$. We have both $p \approx q$ and $p \approxeq q$.

All the laws for strong bisimilarity are still valid. Moreover, Milner's τ-laws hold:

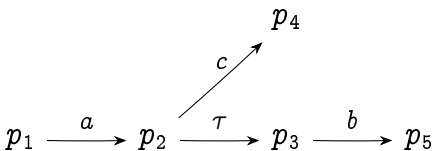
$$p + \tau.p \approxeq \tau.p$$
$$\mu.(p + \tau.q) \approxeq \mu.(p + \tau.q) + \mu.q$$
$$\mu.\tau.p \approxeq \mu.p$$

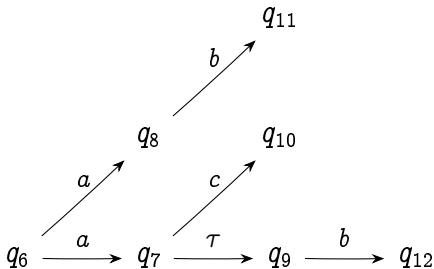
Two Seemingly Equivalent Systems





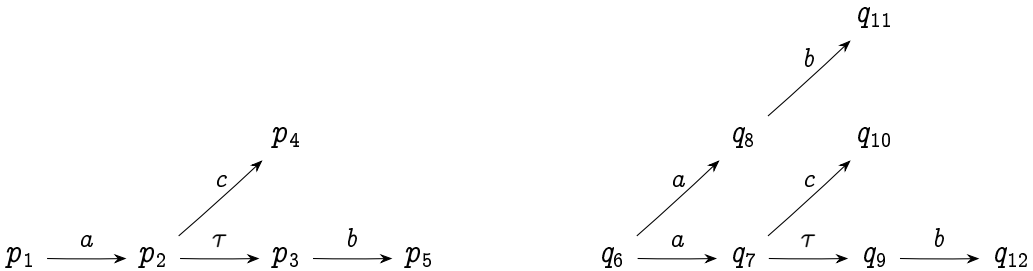
Two Seemingly Equivalent Systems





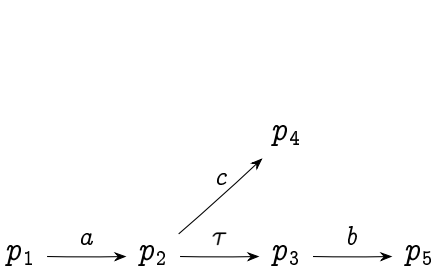
- Processes p_1 and q_6 are weakly bisimilar.

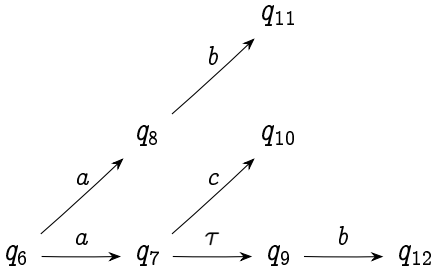
Two Seemingly Equivalent Systems



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- From p_1, the choice between b and c is made after the a-transition.
  From q_6, there is a computation where c is discarded already after a.

Two Seemingly Equivalent Systems





- Processes p_1 and q_6 are weakly bisimilar.
- From p_1, the choice between b and c is made after the a-transition.
  From q_6, there is a computation where c is discarded already after a.
- One may then argue that p_1 and q_6 have a different branching structure (their timing for choices differs) and should not be equivalent.

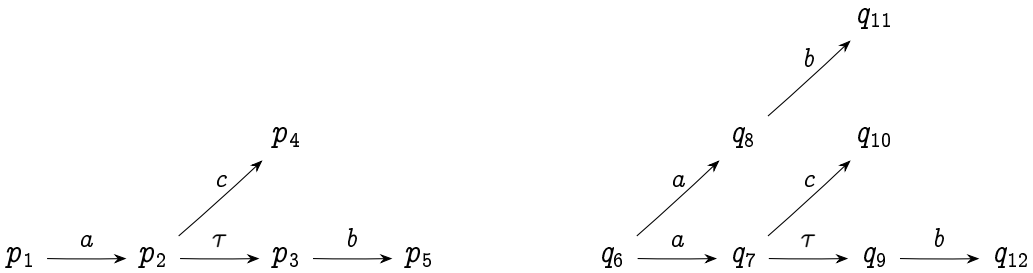
Branching Bisimilarity



$\mathcal{R}$ is a branching bisimulation if and only if $\forall p_1, q_1. (p_1, q_1) \in \mathcal{R}$ we have:

- $\forall p_1'. p_1 \xrightarrow{\lambda} p_1'$,
  - either $\lambda = \tau$ and $\exists q_1'. q_1 \stackrel{\tau}{\Longrightarrow} q_1'$ with $(p_1, q_1') \in \mathcal{R}$ and $(q_1, q_1') \in \mathcal{R}$.
  - or $\exists q, q_1'. q_1 \stackrel{\tau}{\Longrightarrow} q \xrightarrow{\lambda} q_1'$ with $(p_1, q) \in \mathcal{R}$ and $(p_1', q_1') \in \mathcal{R}$.
- (the symmetric conditions from q_1)

Two Seemingly Equivalent Systems



We see that p_1 and q_6 are not branching bisimilar:

- given the transition $q_6 \xrightarrow{a} q_8$, p_1 can respond only with $p_1 \Longrightarrow p_1 \xrightarrow{a} p_2$;
- however, p_2 cannot be related to q_8, as only p_2 can perform c.

Summary



- CCS
- Strong bisimulation and strong bisimilarity ($\sim$)
- Dealing with internal actions: weak bisimilarity ($\approx$)
- Subtle interplay between interal actions and non-determinism
- Weak observational equivalence ($\approx$) and branching bisimilarity

Next: The π-calculus.



The End